

Intellectual Property

Lecture 10

Errors, Failures, and Risks

**Understanding the risks and reasons
for computer failures.**

RISKS digest

- “Forum on Risks to the Public in Computers and Related systems”

BSoD forces students to retake standardized test (May 15, 2007)

2900 Virginia students will have to re-take standardized tests because the computer systems failed during the testing process. There are two descriptions of what went wrong: the testing vendor "reported that there was a problem with a connection between two servers" and students' "computer screens suddenly turned blue and displayed an error message" (i.e., a BSoD). Whether this is one problem or two is unclear... "State officials said there was an unrelated computer problem with online testing last week [where] 1,300 tests were interrupted and that the students will have to be retested."

RISKS digest

Scots Jail hi-tech door locking system broke (September 20, 2005)

Prison officers have been forced to abandon a new security system and return to the use of keys after the cutting-edge technology repeatedly failed. The system, which is thought to have cost over £3 million, used fingerprint recognition to activate the locking system at the high-security Glenochil Prison near Tullibody, Clackmannanshire. ... For more than a month, the 420 inmates - including some murderers and other high-risk inmates - had been able to wander around the high-security jail. Staff claim that the unlimited access to all parts of the prison had allowed some prisoners to settle old scores with rivals.

RISKS digest

Computer problem suspected in erratic Airbus flight (Jan 11, 2008)

An Air Canada flight from Victoria -> Toronto rolled suddenly from side to side and then plunged in the air. It may have suffered technical problems, according to passengers interviewed after the plane was diverted to Calgary. There had been a computer failure and pilots were flying the plane manually. Ten people were admitted to hospital.

What can go wrong?

- Facts about computer errors
 - Error-free software is very, very difficult to achieve.
 - Errors are often caused by several factors.
 - Errors can be reduced by following good software development procedures.
 - Confidence in code can be obtained by keeping it small enough to be understood.

What can go wrong?

- Roles
 - Computer user
 - Should understand limitations of computers
 - Should understand need for proper training and responsible use
 - Computer professional (buyer, developer, system manager...)
 - Needs to understand the sources and consequences of failures
 - Educated member of society
 - Understanding computer risks helps in discussing and deciding public policy.

What can go wrong?

- Categories of computer errors and failures
 - Problems for individuals
 - affects one or a few people
 - System Failures
 - affects large numbers of people or costs large amounts of money or both
 - classic example: telecommunications network
 - Safety-Critical Applications
 - where property may be damaged or destroyed
 - where people may be injured or killed

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What can go wrong?

- Problems for Individuals
 - Billing errors
 - Lack of tests for inconsistencies and inappropriate amounts
- Database accuracy problems
 - Incorrect information resulting in wrongful treatment or acts
- Causes
 - Large, diverse population
 - human common-sense not part of automated processing
 - overconfidence in accuracy of data from a computer
 - information not updated or corrected
 - lack of accountability for errors

What can go wrong?

- System failures
 - Communications
 - Telephone, online, and broadcast services
 - Cellular, paging, long-distance network
 - Broadband outages
 - Business
 - Inventory and management software
 - Financial
 - Stock exchange, brokerages, banks, etc.

What can go wrong?

- Transportation
 - Reservations, ticketing, and baggage handling
- Causes
 - Insufficient testing and debugging time
 - Significant changes in specifications (during and after project start)
 - Overconfidence in system
 - Mismanagement of the project

What can go wrong?

- Safety-Critical Applications

- Uses

- Medicine, health-services
 - Power plants
 - Aircraft
 - Trains
 - Automated Factories
 - Military applications

- Causes of error:

- Overconfidence
 - Insufficient testing
 - Poor HCI
 - Complexity of system (both computer and environment)

Thank You